

JAISWAL, SANJEEV (संजीव जायसवाल)

From: BERA, T.K. (टी.के. बेरा)
Sent: Tuesday, May 31, 2016 3:48 PM
To: Mishra, Rakesh (राकेश मिश्रा)
Cc: Tewari, A.K. (तिवारी, ए. के.); JAISWAL, SANJEEV (संजीव जायसवाल)
Subject: RE: Repeated failure of MP-4 post 18000hrs maintenance
Attachments: FAILURES OF MP-4 AT GAURIDAD.docx

As desired, the detailed analysis report on MP4 failure at Gauridad is attached herewith

Regards

तपस कुमार बेरा / Tapas Kumar Bera

उप महाप्रबन्धक प्रभारी (अनुरक्षण) / DGM.I/C (Maint)

इंडियन ऑयल कॉर्पोरेशन लिमिटेड / Indian Oil Corporation Ltd.

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From: Mishra, Rakesh (राकेश मिश्रा)
Sent: Tuesday, May 31, 2016 1:31 PM
To: BERA, T.K. (टी.के. बेरा)
Cc: Tewari, A.K. (तिवारी, ए. के.)
Subject: FW: Repeated failure of MP-4 post 18000hrs maintenance

Respected Sir,

D(PL) has requested to analyze the below mail and put up a report on the subject.

While preparing the report, instruction/query/comment of D(PL) sent by undersigned through mail dated 24.05.2016 may also be kindly considered.

Regards,
Rakesh

From: Suresh, P. (सुरेश, पी.)
Sent: Saturday, May 28, 2016 1:14 PM
To: BERA, T.K. (टी.के. बेरा)
Cc: Mishra, Rakesh (राकेश मिश्रा); SINHA, JP (सिन्हा जे.पी.); SATI, V. C. (सती, वी. सी.); Tewari, A.K. (तिवारी, ए. के.)
Subject: Repeated failure of MP-4 post 18000hrs maintenance

Sir,

As discussed, detailed analysis of the subject is enclosed .

Regards

P.Suresh
DGM(T&Maint.)
WRPL,Gauridad

cell:7874101073

Detailed analysis on the repeated problems faced on MP-4 Gauridad subsequent to 18000hrs Maintenance

Background:

- MP-4 18000hrs maintenance was started on 31.03.2016
- Maintenance completed on 03.05.2016
- No load trial run done on 04.05.2016
- Load trial run done on 05.05.2016
- Exhaust gas leak was noted on 05.05.2016 which was attended during slave filter removal on 06.05.2016
- New fuel pump adjustment alongwith replacement of fuel filters carried out.
- From 11.05.2016 to 14.05.2016, correction for DE pump bearing lube oil leakage was done

Non availability of MP-4 after major maintenance attributable to the followings

- A. Exhaust gas leakage
- B. Fuel filter availability
- C. Lube oil leakage from pump NDE bearing housing

Problems have been reviewed thoroughly and observations are made based on interaction with WRPL maintenance team

A. REGARDING EXHAUST GAS LEAKAGE:

The exhaust gas leakage occurred from Cylinder no.5 during trial run, which was attended by replacing Metallic Klinger gaskets in exhaust line. Earlier identical problem was observed on 02.07.2014 and 17.08.2015. The flange under reference is the one between the cylinder head and exhaust manifold. Since the cylinder heads are used on rotational basis, every time exhaust is fitted to different heads whenever major maintenance is taken up. In this case, since this is third instance of failure in roughly two years, proper care should have been taken for checking the alignment of exhaust flange with the cylinder head.

Following suggestions has been sent to WRPL

- To check the alignment and probable deformity on the exhaust flange and subsequent correction if required will be done during the next available opportunity.
- Regular check for tightness of exhaust flange will be done during 1500hrs maintenance.
- Bolting sequence for tightening exhaust line to be followed.

B. FUEL FILTER AVILABILITY

Fuel filter is an IC item and Tender for the same was floated well in advance by WRPL but tender got to be cancelled due to technical issue. Order upon fresh tender has been placed and same is under manufacture and first lot will be delivered in two weeks. Hence spare filters were not available at Gauridad. Available limited spare in charge store at different bases were kept as emergency spares. Subsequent to commissioning of motors at Chaksu, the surplus fuel filters from Chaksu was on transit to Gauridad. As the new filter was not available in hand, old filter could not be replaced.

New fuel pump was fitted in MP-4 and running on load problem of fuel pressure low was observed. This problem was partly rectified by doing adjustment in the fuel pump; filter replaced subsequently after receipt from Chaksu and problem was fully rectified.

C. LUBE OIL LEAKAGE FROM PUMP NDE BEARING HOUSING:

- The problem of lube oil leakage from pump DE bearing housing was known before hand and was planned to be attended during the major maintenance.
- Cyclone separator cleaning, Bearing check and replacement of Deflector plates of pump DE bearing housing were taken up.
- The engine load trial was done on 05.05.2016 and no lube oil leakage was noticed.
- On 10.05.2016, while attending the fuel pressure problem, the lube oil seepage through deflector plate of DE bearing housing were checked and grub screw tightened
- On 11.05.2016 nightshift, since the lube oil leakage was observed MP-4 was stopped.
- On 12.05.2016 to 14.05.2016, the deflector plates were machined to match the slots and refitted.
- Generally the deflector plates are same for both NDE and DE side in all pumps, but in this case the slots were found to be different from DE side and hence the machining had to be taken up.
- Even now there is slight drop of leakage in NDE side. But compared to the oil pool available in the reservoir, the machine can be safely run with observation.
- There could be two reasons for such leakage.
 - One is improper fit of deflector plates.
 - The other is shaft diameter deformity.
- Through the aforementioned exercise all possible corrections have been done for the first cause (improper fit of the deflector plates). Now for the second probable cause. Dismantling of the pump shaft is required for precisely measure and locate the deformity and re-building of the shaft.
- The other MLPU, MP-3 Gauridad was not available for prolonged periods due to deformity in the liner landing area of cylinder -1. This has been temporarily resolved and the pump load trial is being done on 28.05.2016.
- Both MP-3 and MP-4 are to be taken up for corrections in the coming days. Since there is pressure on throughput side, the availability of the Gauridad MLPUs is critical at this stage. Once Surendranagar to Viramagam section of DBL is also commissioned, there will be sufficient flexibility in operation and MP-4 can be taken up for correction. MP-3 being more expensive and intricate work, would need time for correction. It is proposed to take this up after MP-4 of Vadinar is shifted and installed at Gauridad.

To summarise,

- There are certain short comings in the execution of said maintenance as well as the personnel engaged handling the works, though none of them eligible to be termed as wilful negligence. However, this needs correction during course of execution of the works. It has been realised and appropriate measures like nominating a person responsible for looking into these during major or breakdown maintenance has already been taken.
- For pump deflector, there could be two probable causes for the problem, out of which attention was paid to the most common cause i.e. replacement of deflector plate. Second reason could be inherent mismatch between baffle plate slots, which the team WRPL may not be aware of or had escaped their attention. The focus was on the DE side baffle plate and it was attended but the NDE site baffle plate was not checked thoroughly else the machining of the shaft could have been taken up during major maintenance. In order to resolve the problem completely, the second probable cause needs to be attended in the next possible opportunity; however this requires significant outage of pump.

- Checking of tightness of exhaust manifold bolts to be ensured in every 1500 hrs maintenance.
- While preparing the list of defects to be attended during major maintenance, all failures from the date of completion of last major maintenance will be taken into account.